

An Integrated AI- and IoT-Based Smart Water Management Framework for Reservoir-Fed Irrigation Agriculture in Sri Lanka

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Abstract—Sri Lanka's reservoir-fed irrigation schemes supply water to a large share of the country's paddy and vegetable command areas under administrative quota allocations. The decisions that shape daily scheme operation—reservoir release volumes, crop choice, field-valve timing, and crop-stress response—are still taken by separate actors with limited data exchange between them. This paper presents an integrated AI- and IoT-based smart water management framework, implemented as the Adaptive Smart Irrigation and Crop Optimization Platform (ASICOP), that brings these four decision streams into a single feedback loop for the Udawalawe scheme. The framework has four linked modules: reservoir forecasting through a multi-layer ensemble spanning linear regression, ARIMA/SARIMA, Random Forest, Gradient Boosting, LSTM, and quantile regression; adaptive crop and area optimization combining a Random Forest recommender, a neural-network/LightGBM price predictor, a Fuzzy-TOPSIS suitability scorer, and a water-aware area allocator; field-level smart irrigation using a HistGradientBoostingRegressor for next-day canal release and a Random Forest valve controller driven by IoT telemetry; and hybrid crop-health monitoring combining a Sentinel-2 NDVI/NDWI Random Forest zone classifier with a MobileNetV2 PlantVillage disease classifier. On verified test data, the release regressor attains $R^2 = 0.7949$ (RMSE = 0.7108 MCM/day) on 1,095 held-out days from a 32-year Udawalawe hydrological record; the disease classifier reaches 95.43% validation accuracy across 38 classes; the crop recommender reaches 34.07% five-class test accuracy (macro-F1 = 0.32), fourteen points above the 20% random baseline; and the price predictor reaches $R^2 = 0.167$ under five-fold cross-validation. Modules communicate through Redis-cached service calls and an MQTT event bus. The paper's main contribution is to show that reservoir-scale forecasting, field-level IoT actuation, satellite-derived crop-health feedback, and quota-aware crop planning can be designed as one coherent, loosely coupled ecosystem for Sri Lankan reservoir-fed agriculture.

Keywords—adaptive irrigation, crop health monitoring, decision-support systems, Fuzzy-TOPSIS, Internet of Things (IoT), machine-learning forecasting, remote sensing, reservoir operation, smart water management, Sri Lanka.

I. INTRODUCTION

Agriculture still accounts for the largest share of global freshwater withdrawals, and the pressure on that share is rising as populations grow, yields must intensify, and rainfall patterns shift under a changing climate. In Sri Lanka, where paddy and field-crop farming together underpin a substantial portion of rural livelihoods, most irrigated land is served by an engineered network of

reservoirs—locally called wewas or tanks—that impound seasonal rainfall and release it through canal systems under administrative quota allocations.

The Udawalawe scheme, built around the Udawalawe reservoir on the Walawe Ganga in the island's dry zone, is one of the larger of these systems. It supplies water across two cropping seasons each year and serves thousands of smallholder farmers under quotas issued by the Irrigation Department and the Mahaweli Authority.

Running a scheme of this size requires four distinct kinds of decision, each currently handled by a different actor. The reservoir operator decides how much water to release each day. Agricultural officers and farmers decide which crops to plant and on how much land. Field operators decide when to open or close individual field valves. Extension staff monitor crop health and respond to disease or water-stress symptoms. In practice, these four streams rarely exchange data. The result is a decision loop that is open where it should be closed.

Advances in machine learning, satellite remote sensing, and the Internet of Things now make it possible to link these four decisions. We present such a framework and describe its implementation as the Adaptive Smart Irrigation and Crop Optimization Platform (ASICOP). ASICOP organises four tightly linked modules behind a common API gateway: (F1) IoT smart water management and field-level irrigation control; (F2) hybrid satellite- and image-based crop health and water-stress detection; (F3) machine-learning time-series forecasting of reservoir levels and releases with quantile risk bands; and (F4) adaptive crop and area optimization.

This paper makes four specific claims. First, a 32-year record of Udawalawe hydrological data trains a HistGradientBoostingRegressor that predicts next-day canal release with $R^2 = 0.7949$ and RMSE = 0.7108 MCM/day on a held-out 2023–2025 test set. Second, a MobileNetV2 classifier fine-tuned on the PlantVillage dataset (~54,306 images, 38 classes) attains 95.43% validation accuracy. Third, a Random Forest crop recommender coupled with a Fuzzy-TOPSIS suitability layer produces quota-aware crop plans. Fourth, these modules are designed and deployed as a single loosely coupled integration with explicit cross-service event flows.

II. LITERATURE REVIEW

A. Reservoir and Hydrological Forecasting

Classical reservoir operation has long relied on physically based models—SWAT, HEC-ResSim, and Budyko-family water-balance formulations—calibrated on decadal streamflow records [9], [10]. Data-driven alternatives have been building in parallel. Random Forests [5] and gradient-boosted ensembles [8] have repeatedly outperformed linear baselines on daily and weekly

inflow prediction when sufficient historical data is available. Recurrent architectures—most notably LSTM and its variants [6]—have been applied to longer forecast horizons [11]. Probabilistic outputs such as quantile regression have become particularly important, where P10/P50/P90 spread is more actionable than a single point estimate [26].

B. Smart Field Irrigation and IoT Actuation

Field-level smart irrigation has been productive for over a decade. ESP32 and Arduino-class microcontrollers, combined with low-cost capacitive moisture sensors and MQTT publishing, have made such systems inexpensive enough for smallholder deployment [13]. Two limitations recur: most field-scale systems are disconnected from scheme-scale decisions, and most classifier-based valve controllers are evaluated on synthetic or vendor-collected data with simple label rules.

C. Satellite- and Image-Based Crop Health Monitoring

The remote-sensing literature has produced two mature tracks. The first uses multispectral satellite indices—NDVI, NDWI, NDRE—to classify zones as healthy, mildly stressed, or severely stressed [18]. Sentinel-2, at 10-metre spatial resolution and five-day revisit, has made this practical at the field scale [12]. The second track uses convolutional neural networks to classify diseases from leaf images. The PlantVillage dataset [1] has become the standard benchmark, with ~54,306 images across 38 class labels. MobileNetV2 is a frequent choice because its inverted-residual architecture keeps inference latency low for on-device deployment [2].

D. Crop Recommendation, Suitability, and Area Optimization

Crop choice and area allocation under water and land constraints is a classical operations-research problem. More recent work has added multi-criteria decision-making methods—Fuzzy-TOPSIS—to blend agronomic suitability, market attractiveness, and risk tolerance into a single ranking [3]. Machine-learning crop recommenders, typically Random Forest [5] trained on soil-and-climate feature vectors, have become common in precision-agriculture platforms [20], [21]. ASICOP's F4 module integrates reservoir quantile forecasts as allocation constraints—a linkage rare in prior work.

E. Integrated Systems and Positioning of ASICOP

Table I contrasts representative prior work with ASICOP across seven axes. Most existing integrated platforms couple two technical streams rather than four. ASICOP is the only system covering all four streams—reservoir forecasting, field IoT, satellite and image-based crop health, and crop planning—under a quota-based Sri Lankan scheme, with explicit cross-module event flows.

TABLE I. COMPARISON OF REPRESENTATIVE PRIOR SYSTEMS WITH ASICOP

System	Reservoir	Field IoT	Sat. Health	Image Health	Crop Opt.
Prior Work [Reservoir ML]	✓	—	—	—	—
Prior Work [Smart IoT]	—	✓	—	—	—
Prior Work [Sentinel-2]	—	—	✓	—	—
Prior Work [PlantVillage]	—	—	—	✓	—
Prior Work [IoT+Sat]	—	✓	✓	—	—
Prior Work [Res.+Crop]	✓	—	—	—	✓
ASICOP (this paper)	✓	✓	✓	✓	✓

III. RESEARCH GAP

Three specific gaps emerge from the review. First, each of the four technical streams has a mature stand-alone literature, but their integration under a single decision-support framework is rare—most platforms couple two streams rather than four. Second, prior machine-learning work on reservoir forecasting, smart IoT irrigation, satellite stress monitoring, and crop optimisation disproportionately targets large schemes in data-rich regions; Sri Lankan reservoir-fed schemes under administrative quota allocation are under-represented. Third, even where integration is attempted, the mechanism is usually thin—a shared dashboard or a one-directional feed. ASICOP addresses all three gaps.

IV. RESEARCH PROBLEM

In a reservoir-fed quota-based irrigation scheme such as Udawalawe, four interdependent decisions—(i) how much water the reservoir should release each day, (ii) which crops farmers should plant and on how much land, (iii) when individual field valves should open in response to moisture, rainfall, and stress conditions, and (iv) how to detect and respond to crop-health anomalies—are currently taken by different actors on different time scales with limited data exchange. The research question is whether a single AI- and IoT-based framework can link these four decisions into one coherent feedback loop at the scheme scale, delivering per-module performance comparable to prior work while providing the integration layer that prior work has not.

V. OBJECTIVES

The main objective is to design, implement, and evaluate an integrated AI- and IoT-based smart water management framework that links reservoir-level forecasting, crop and area planning, field-level irrigation control, and crop-health monitoring into a single feedback loop for reservoir-fed quota-based irrigation schemes in Sri Lanka, with ASICOP as the reference implementation.

Six specific objectives decompose the main objective:

- SO1 (F3) Develop a multi-layer forecasting stack producing P10/P50/P90 quantile risk bands for daily reservoir water level and release.
- SO2 (F1) Design a field-level IoT telemetry, decision, and actuation pipeline using a HistGradientBoostingRegressor for next-day canal release and a Random Forest valve classifier.
- SO3 (F2) Develop a two-track crop-health pipeline using Sentinel-2 NDVI/NDWI for zone-level stress maps and MobileNetV2 for point-of-care disease classification.
- SO4 (F4) Build a quota-aware crop-planning stack combining a Random Forest recommender, LightGBM price predictor, Fuzzy-TOPSIS suitability scorer, and water-aware area allocator.

SO5 Design and implement the cross-service integration layer linking F1–F4 through an MQTT event bus and Redis-cached REST service calls.

SO6 Evaluate each module on verified test data and document current limitations with explicit remediation pathways.

VI. CONTRIBUTIONS

This paper makes six contributions:

1. An integrated reservoir-to-field-to-feedback framework for Sri Lankan quota-based irrigation schemes, composing all four decision streams with explicit MQTT event flows and Redis-cached graceful degradation.
2. A 32-year Udawalawe release-prediction model (HistGradientBoostingRegressor, $R^2 = 0.7949$, RMSE =

0.7108 MCM/day) with 46 engineered features outperforming Ridge and Random Forest baselines.

3. A hybrid satellite- and image-based crop-health pipeline combining a Random Forest Sentinel-2 NDVI/NDWI classifier with a MobileNetV2 disease classifier achieving 95.43% validation accuracy on PlantVillage.
4. A quota-aware adaptive crop and area optimization stack delivering 34.07% five-class test accuracy (+14 pp over random) with Fuzzy-TOPSIS suitability scoring and P10/P50/P90-constrained area allocation.
5. A five-layer probabilistic forecasting stack producing P10/P50/P90 reservoir-level forecasts with unsupervised anomaly detection for drought and flood alerting.
6. Transparent documentation of eight specific limitations each paired with a concrete remediation pathway, establishing a clear bridge to operational deployment.
7. A consistent six-block methodology pattern (inputs → preprocessing → features → model → outputs → integration) applied uniformly across all four modules, making the modules comparable, the cross-module dependencies explicit, and the methodology reusable by researchers working on other Sri Lankan schemes.

VII. PROPOSED SYSTEM ARCHITECTURE

ASICOP is structured as a microservice architecture deployed behind a shared API gateway. The design separates concerns into three planes—client, service, and data—and connects the four functional modules through deliberate cross-module event flows.

A. Client Plane

Three classes of client interact with the platform. A Progressive Web Application (PWA) serves the primary interface for reservoir operators, agricultural officers, and scheme managers, delivering forecast dashboards, crop-plan outputs, and zone health maps. A farmer-facing mobile interface provides field-level valve status, irrigation advisories, and crop health alerts. ESP32-class microcontrollers in the field stream sensor telemetry—soil moisture, air temperature, local rainfall, and field-level water height—over the MQTT event bus and receive valve-actuation commands from the irrigation service.

B. Service Plane

Seven services sit behind an API gateway on port 8000. Table II lists each service and its primary responsibility.

TABLE II. ASICOP SERVICE PLANE

Service	Port	Primary Responsibility
Authentication	8001	JWT-based authentication and user management
F1 — IoT Irrigation	8002	Canal-release forecasting; field-valve actuation decisions
F3 — Forecasting	8003	Reservoir level and rainfall forecasting; P10/P50/P90; alerts
F4 — Optimization	8004	Crop recommendation; price prediction; Fuzzy-TOPSIS; area allocation
IoT Broker	8006	MQTT routing between ESP32 field devices and irrigation service
F2 — Crop Health	8007	Sentinel-2 NDVI/NDWI zonal stress analysis; disease classification

C. Data Plane

A PostgreSQL relational database persists time-series hydrological records, crop plans, valve-actuation logs, and user data. Redis provides cross-service result caching with a ten-minute

TTL. Mosquitto brokers the MQTT event bus. A Prometheus and Grafana layer provide operational observability.

D. Cross-Module Event Flows

Four integration dependencies constitute the feedback loop that distinguishes ASICOP from a collection of disjoint subsystems:

F3 → F1 (Rainfall suppression). When F3 predicts rainfall exceeding 5 mm in 24 hours, it publishes a forecast. update.v1 MQTT event. The F1 service defers pending valve-open decisions for that scheduling cycle.

F2 → F1 (Stress-based priority elevation). The F2 service publishes crop. stress.v1 events when a zone's stress index exceeds a threshold. F1 retrieves the stress_penalty_factor via Redis-cached REST and elevates irrigation priority proportionally.

F2 → F4 (Suitability penalisation). Before Fuzzy-TOPSIS scoring, F4 reads zone-level stress summaries and applies: $\text{effective_suitability} = \text{base_suitability} \times (1 - \text{penalty_factor})$.

F4 → F3 (Quantile-constrained allocation). The F4 area allocator reads P10/P50/P90 release forecasts from F3. P50 sets the primary water-availability bound; P10 sets the conservative floor constraint.

On upstream failure, each service falls back to its Redis-cached last-known-good output, preventing a single service outage from cascading across the platform.

TABLE III. CROSS-MODULE DATA FLOWS

Producer	Consumer	Data Passed	Mechanism	Effect
F3	F1	Rainfall >5mm flag	MQTT forecast.update.v1	Suppresses valve-open
F2	F1	stress_penalty_factor	Redis-cached REST	Elevates irrigation priority
F2	F4	stress_penalty_factor	Redis-cached REST	Penalises TOPSIS suitability
F3	F4	P10/P50/P90 release	Redis-cached REST	Bounds area allocation

VIII. METHODOLOGY

Each module below follows the same six-block pattern: Inputs → Preprocessing → Features → Model → Outputs → Integration.

A. F1 - IoT Smart Water Management and Field Irrigation Control

Inputs. The primary source is the Udawalawe Hydrological Dataset from the Irrigation Department of Sri Lanka: an Excel workbook of 32 annual sheets (1 January 1994 to 31 December 2025) with daily records of reservoir water level (mMSL), rainfall (mm), inflow (MCM), canal releases (MCM), and evaporation. A secondary input is field-level IoT telemetry from ESP32 microcontrollers over MQTT, currently represented by 1,000 synthetic samples (seed=42).

Preprocessing. All 32 annual sheets are concatenated into a single DataFrame of 11,687 raw rows. After removing non-numeric entries, rows with multiple missing signals, and instrument-failure outliers, 10,945 cleaned daily records remain. The dataset is split chronologically: 9,850 rows (1994–2022) train and 1,095 rows (2023–2025) test.

Features. 46 input features are engineered: ten base hydrological signals, twenty lag features (lag-1/2/3/7 over five signals), nine

rolling-mean features (windows 3, 7, 14 days), and three calendar variables (month, day-of-year, binary season indicator). Feature importance analysis confirms that lag-1 and lag-2 water-level features dominate the HGB prediction, consistent with the strong autoregressive structure of reservoir levels. Seasonal calendar features contribute the next-largest share, reflecting the bimodal Maha/Yala agricultural season pattern of the Walawe basin.

Model. Three regression models are evaluated: Ridge Regression, Random Forest [5] (100 estimators), and HistGradientBoostingRegressor (max_depth=6, learning_rate=0.07, max_iter=250). A separate RandomForestClassifier [5] (n_estimators=10) is trained on 1,000 synthetic field-sensor samples to produce valve actuation decisions: OPEN, CLOSE, or EMERGENCY_RELEASE (triggered at 95 mMSL; 80 mMSL minimum-safe-level).

Outputs & Integration. The HGB regressor publishes a scalar next-day canal-release forecast as a reservoir.release.v1 MQTT event cached in Redis. When F3 issues a rainfall flag, F1 defers valve-open commands. When F2 publishes a stress event, F1 raises irrigation priority. The HGB forecast also feeds the F4 area allocator as a water-availability component.

B. F2 - Hybrid Satellite and Image-Based Crop Health Monitoring

Inputs. Track A: Sentinel-2 Level-2A imagery for the Udawalawe scheme perimeter via Google Earth Engine (GEE project fyp-geesrilanka, COPENICUS/S2_SR catalogue [4]). A 2023 median composite at 10-metre resolution is derived within a 15 km buffer centred on (6.42°N, 80.89°E). Track B: PlantVillage dataset [1] comprising ~54,306 RGB leaf images across 38 disease-class labels covering 14 crop species.

Preprocessing. Track A: five-stage vegetation validation (cloud masking, water-body exclusion, urban exclusion, coverage check, NDVI/NDWI computation). A stratified sample of 5,000 pixels is assigned proxy labels—healthy, mild stress, severe stress—based on published NDVI/NDWI thresholds. Track B: images resized to 224×224, normalised to ImageNet statistics, split 80/20 with class-stratified sampling.

Model. Track A: a Random Forest classifier [5] trained on 5,000 pixels attains ~99% accuracy (proxy-label evaluation; reflects rule-learning, not agronomic ground truth). Track B: MobileNetV2 [2] backbone (pre-trained on ImageNet, frozen) with a GlobalAveragePooling2D → Dense(128, ReLU) → Dense(38, Softmax) head (~49,190 trainable parameters) trained for 10 epochs, batch size 32, Adam lr=0.001.

Outputs & Integration. Track A produces a zone-level health map (healthy / mild stress / severe stress). Track B produces a 38-class disease label and confidence. The stress index is computed as: stress_index = min(1.0, (mild_ratio × 0.5) + severe_ratio). F2 publishes crop.stress.v1 events; F1 and F4 consume the stress_penalty_factor via Redis-cached REST.

C. F3 - ML Time-Series Forecasting and Alerting

Inputs. Same Udawalawe Hydrological Dataset as F1 (32 sheets), supplemented by live weather data from the Open-Meteo API [23] for the Udawalawe catchment coordinates. The production service iterates all 32 sheets; the notebook prototype inadvertently loads only 1994 (see Limitation L4 in Section XI).

Preprocessing. In the production service, the 32-sheet concatenation follows the same pipeline as F1. An additional feature-alignment step resamples and joins the Open-Meteo weather forecast horizon to the daily hydrological cadence. In the notebook prototype, the 365-row 1994 subset is cleaned to 358 rows and split as 286 training / 72 test.

Features. Lagged water-level and rainfall values (lag-1/2/3/7), rolling means (7-day and 30-day), and calendar variables. Feature-importance analysis identifies water_level_lag_1 as the dominant predictor at 49.3% total importance.

Model — Five-Layer Stack. Layer 1: Ridge Regression baseline. Layer 2: ARIMA/SARIMA with AIC-selected orders. Layer 3: Random Forest [5], Gradient Boosting [8], and LSTM [6] (LSTM(64) → Dropout(0.2) → LSTM(32) → Dropout(0.2) → Dense(16) → Dense(1); 32,673 parameters). Layer 4: weighted ensemble aggregator fitted on the validation partition. Layer 5: unsupervised anomaly detection (Isolation Forest, LOF, DBSCAN, Z-score; majority vote). Three quantile Gradient Boosting models [8] at $\alpha = 0.10, 0.50, 0.90$ produce the P10/P50/P90 bands.

Outputs & Integration. Daily point forecasts of reservoir water level and canal release, P10/P50/P90 quantile bands, and drought/flood alert flags when P50 breaches 80 mMSL or 95 mMSL. F3 publishes forecast.update.v1 events; F4 consumes P10/P50/P90 as area-allocation constraints.

D. F4 - Adaptive Crop and Area Optimization

Inputs. Primary: Hector Kobbekaduwa Agrarian Research and Training Institute Government Retail Price dataset [24] (71,737 weekly records, five vegetable crops, 37 Sri Lankan districts, 2015–2024). Supplementary: Sri Lanka Climate dataset (314,000 daily records), Paddy Cultivation dataset (1,039 records), and Rice Time Series dataset (324 monthly records).

Preprocessing. District names harmonised across Hector and climate datasets. Missing price values imputed with district-level crop-month means. Dataset split under five-fold cross-validation for the price predictor; separate holdout for the crop recommender.

Features. The LightGBM production price predictor [7] is designed around 24 features: spatial (district, zone identifier), temporal (month, year, season indicator), weather (mean temperature, cumulative rainfall, mean humidity), crop identity (label-encoded), lagged price at lag-1/2/3 months, and three derived indicators (price volatility, rainfall deviation from district mean, growing-season binary flag). The Random Forest crop recommender [5] and PyTorch neural-network prototype operate on the temporal, district, and crop-identity subset, as soil-type and nutrient features are unavailable in the market-price record.

Model - Four Components. (1) RandomForestClassifier [5] (100 estimators) maps district-season-weather to one of five crop categories. (2) PricePredictorNN (PyTorch): dual-embedding network with separate crop-identity and district embeddings, two fully connected layers (5,409 parameters total). (3) Fuzzy-TOPSIS [3] with five criteria—soil suitability, water coverage, yield, water sensitivity, growth duration—and weights [0.25, 0.25, 0.20, 0.15, 0.15], with trapezoidal fuzzy numbers and stress-adjusted effective suitability. (4) Greedy area allocator ranks crops by Rs/mm efficiency; P50 is the primary water quota bound, P10 sets the conservative floor.

Outputs & Integration. Ranked crop-suitability table per zone, quota-aware area allocation plan, and expected-revenue estimates under P10/P50/P90 scenarios. F4 reads F3 quantile forecasts and F2 stress summaries before each planning cycle.

IX. EXPERIMENTAL SETUP

Table IV summarises the dataset used by each module, its size, provenance, and the train/test split applied.

TABLE IV. MODULE DATASETS AND PARTITIONS

Module	Dataset	Records	Split
F1 Release Regressor	Udawalawe Hydrol. 1994–2025 (32 sheets)	10,945 clean	Chronological: 9,850 train / 1,095 test
F1 Valve Classifier	Synthetic field-sensor (seed=42)	1,000 samples	80/20 random split
F2 RF Satellite	Sentinel-2 2023 composite (GEE)	5,000 pixels, 3 classes	80/20 stratified split
F2 MobileNetV2	PlantVillage (38 classes)	~54,306 images	80/20 class-stratified
F3 Notebook Proto.	Udawalawe 1994 only (single-sheet bug)	358 clean	286 train / 72 test
F4 Crop Recommender	Hector Govt Retail Prices 2015–2024	71,737 records, 5 crops	Holdout test partition
F4 Price Predictor	Hector + Sri Lanka Climate merged	71,737 + climate rows	5-fold cross-validation

Each module's primary model is compared against at least one baseline, as summarised in Table V.

TABLE V. PER-MODULE BASELINES

Module	Primary Model	Baseline(s)
F1 — Release regressor	HistGradientBoostingRegressor	Ridge Regression; Random Forest [5]
F1 — Valve classifier	RandomForestClassifier [5]	Rule-based threshold (moisture <30% AND temp >25°C)
F2 — RF satellite	Random Forest [5] (proxy labels)	NDVI/NDWI threshold rule (direct)
F2 — MobileNetV2	MobileNetV2 fine-tuned [2]	Random chance (1/38 ≈ 2.6%)
F3 — Forecasting	Gradient Boosting [8] (best of three)	Linear Regression; ARIMA/SARIMA
F4 — Crop recommender	RandomForestClassifier [5]	Random baseline (1/5 = 20%)
F4 — Price predictor	PricePredictorNN (PyTorch)	RF log-scale [5] (R^2 0.486); Linear Reg. (R^2 0.044)

All results are labelled with one of four status tags: [MEASURED] obtained on real-world held-out data; [SYNTHETIC-ONLY] on synthetic or proxy-labelled data; [PROTOTYPE / LIMITATION] on real data under a known methodological constraint; [PROJECTED / NOT YET RUN] expected but not yet measured.

X. RESULTS AND DISCUSSION

A. F1 - Canal-Release Forecasting [MEASURED]

Table V presents test-set performance of the three regression models on the 2023–2025 chronological holdout ($n = 1,095$).

TABLE V. F1 RELEASE REGRESSOR COMPARISON (N = 1,095)

Model	MAE (MCM/day)	RMSE (MCM/day)
Ridge Regression (baseline)	0.5191	0.7497
Random Forest (baseline) [5]	0.4366	0.8514
HistGradBoostingReg. (selected)	0.4412	0.7108

The HistGradientBoostingRegressor attains the best RMSE (0.7108 MCM/day) and highest R^2 (0.7949). RMSE was the primary selection criterion because underestimating a large release event carries greater operational risk than minor day-to-day

forecast noise. The R^2 of 0.7949 indicates the model explains approximately 79.5% of the variance in next-day canal release from a 46-feature input spanning multiple drought and flood cycles.

The F1 valve classifier is trained and evaluated on 1,000 synthetic samples [SYNTHETIC-ONLY]. Field-calibrated retraining using real ESP32 telemetry is required before this component can be considered production-ready.

B. F2 — Crop Health Monitoring [MEASURED / PROXY-LABEL]

Table VI reports per-epoch validation metrics for the 10-epoch MobileNetV2 PlantVillage training.

TABLE VI. F2 MOBILENETV2 VALIDATION HISTORY (SELECTED EPOCHS)

Epoch	Val Accuracy (%)	Val Loss	Notes
1	~85.0	—	Initial convergence
4	~95.05	0.1452 (best loss)	Best validation loss
9	95.43 (best acc.)	—	Best validation accuracy
10	~95.1	—	Final epoch

The MobileNetV2 classifier [2] reaches 95.43% validation accuracy at epoch 9 across 38 PlantVillage classes [1], with ~49,190 trainable parameters and ~5.8 hours total training. The gap between best-accuracy (epoch 9) and best-loss (epoch 4) checkpoints indicates mild overfitting in the final epochs. Two limitations qualify this result: (i) PlantVillage images are taken under controlled conditions, so field accuracy on smallholder photographs is expected to be lower; (ii) paddy (rice)—the dominant Udawalawe crop—has no class in PlantVillage.

The Random Forest satellite classifier [5] attains ~99% accuracy on its held-out pixel subset [PROXY-LABEL]. This reflects rule-learning from the NDVI/NDWI threshold labels, not agronomic ground truth.

C. F3 - Forecasting [PROTOTYPE / LIMITATION]

Table VII reports performance on the 1994-only test partition ($n = 72$). All results are affected by the single-sheet loading bug.

TABLE VII. F3 NOTEBOOK COMPARISON (1994-ONLY SUBSET, N = 72)

Model	RMSE (mMSL)	MAE (mMSL)	R^2
Gradient Boosting [8]	2.8763	2.7027	−2.2912
Random Forest [5]	3.1813	3.0068	−3.0260
LSTM [6] (32,673 params)	3.6707	3.2944	−17.5414
P80 quantile coverage	—	—	8.33% (target ≥70%)

Negative R^2 values arise because training on 286 days from a single hydrological year cannot capture inter-annual variance patterns. This is not evidence of model inadequacy but a direct consequence of the data-loading error. Based on the F1 module's experience with the same full dataset, projected multi-sheet performance is $R^2 > 0.80$, RMSE < 1.0 mMSL, and P80 coverage > 70% [PROJECTED / NOT YET RUN].

D. F4 - Adaptive Crop and Area Optimization [MEASURED]

The RandomForestClassifier attains 34.07% test accuracy and 41.14% training accuracy on the five-class crop-recommendation task, with macro-F1 = 0.32. Best per-class F1 is 0.40 for CARROT (recall 0.72); LEEKS achieves the highest precision at 0.48. The 34.07% figure is +14 percentage points above the 20% five-class random baseline.

TABLE VIII. F4 PRICE PREDICTION MODEL COMPARISON

Model	R ²	MAE (Rs/kg)	RMSE (Rs/kg)
Linear Regression	0.044	—	—
RF log-scale (baseline) [5]	0.486	—	—
PricePredictorNN (PyTorch)	0.167	115.66	175.81
LightGBM [7] (24 features)	Not yet run	Not yet run	Not yet run

The PricePredictorNN achieves test $R^2 = 0.167$, with five-fold cross-validation confirming stable generalisation (mean $R^2 = 0.155 \pm 0.009$, train/validation ratio 1.01). The RF log-scale baseline achieves $R^2 = 0.486$, reflecting its ability to fit the non-linear log-price distribution; the neural network's dual-embedding design is the more principled approach for out-of-sample district generalisation.

E. Cross-Module Integration

All four cross-module data flows are implemented and functional in the prototype deployment: F3 rainfall events suppress F1 valve-open commands; F2 stress events elevate F1 irrigation priority; F2 stress penalties modify F4 Fuzzy-TOPSIS scores; and F3 P10/P50/P90 quantiles constrain the F4 area allocator. All four flows have been verified through manual inspection of service logs and Redis cache state.

No end-to-end experiment has been run measuring the effect of integration on aggregate water use or yield [PROJECTED / NOT YET RUN]. This is the primary gap between the architectural contribution and a full empirical validation, and is the central task of the forthcoming deployment study.

XI. LIMITATIONS

This paper documents eight specific limitations, each with its root cause and remediation pathway.

L1 — F1 Synthetic-Only Valve Training. The RandomForestClassifier for field-valve actuation is trained on 1,000 synthetic sensor samples from a simplified two-condition rule. It does not capture crop-growth-stage variation, inter-field soil heterogeneity, or real sensor noise. The expected synthetic-distribution accuracy (>95%) cannot be extrapolated to real field conditions.

L2 — F2 MobileNetV2: No Paddy Class. Paddy (rice) is not among the 14 PlantVillage species. The MobileNetV2 classifier's degree of generalisation to paddy leaf images is unknown and unvalidated.

L3 — F2 RF Satellite: Proxy-Derived Labels. The ~99% accuracy reflects model learning of the NDVI/NDWI threshold rule, not independent agreement with agronomic ground truth.

L4 — F3 Single-Sheet Loading Bug. The F3 notebook reads only the 1994 sheet due to a pd.read_excel call without sheet iteration. All negative R^2 values in Table VII are direct consequences of this data-loading error.

L5 — F4 Greedy Area Allocator. The active allocator uses greedy Rs/mm heuristic and cannot enforce inter-crop rotation constraints or multi-season planning. The PuLP MIP formulation is coded but not wired as the active solver.

L6 — F4 Hector Dataset Excludes Paddy. Paddy price modelling relies on a separate 324-record monthly Rice Time Series, substantially smaller than the weekly Hector records.

L7 — Quota Enforcement Is Advisory. ASICOP provides decision support within the existing administrative quota framework. It cannot physically enforce compliance.

L8 — Udawalawe-Specific Calibration. All numerical thresholds (80 mMSL minimum-safe, 95 mMSL emergency, 30% moisture threshold, Fuzzy-TOPSIS weights) are calibrated to the Udawalawe scheme and require re-parameterisation for other Sri Lankan schemes.

XII. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper has presented ASICOP, an integrated AI- and IoT-based smart water management framework designed for reservoir-fed quota-based irrigation schemes in Sri Lanka. The framework links four functional modules through a microservice architecture whose cross-module event flows, Redis-cached service calls, and graceful-degradation fallbacks keep four decision streams in continuous, operationally coherent communication.

Per-module results on verified test partitions establish four concrete technical outcomes. The HistGradientBoostingRegressor achieves $R^2 = 0.7949$ (RMSE = 0.7108 MCM/day) on 1,095 held-out days from a 32-year record. The MobileNetV2 disease classifier achieves 95.43% validation accuracy on 38-class PlantVillage, while the paddy-class gap and controlled-imaging conditions remain primary barriers to direct field deployment. The F4 Random Forest crop recommender achieves 34.07% test accuracy (+14 pp over random), with the Fuzzy-TOPSIS layer providing agronomic suitability ranking that the price-only classifier cannot. The F3 prototype yields negative R^2 on all three supervised models—not evidence of modelling failure but a direct consequence of the inadvertent single-sheet data-loading error.

The integration contribution rests on the architectural design: four cross-module event flows that reflect how a well-informed scheme operator would link these four decisions. Whether the design delivers measurable improvements in water use efficiency, crop yield, or decision quality over the fragmented status quo is the open empirical question that the forthcoming deployment study must answer.

B. Future Work

R1 — F3 Multi-Sheet Re-run. Correcting the pd.read_excel call to iterate all 32 sheets and re-running all models is the single highest-impact next action, requiring no new data collection. Projected outcomes: $R^2 > 0.80$, RMSE < 1.0 mMSL, P80 coverage > 70%.

R2 — Field-Calibrated F1 Valve Retraining. Deploying ESP32 sensors for 6–12 months in instrumented Udawalawe fields to collect real soil-moisture, temperature, and valve-state telemetry for RF classifier retraining.

R3 — Paddy-Class Augmentation for F2. Collecting a paddy-specific leaf image dataset covering bacterial leaf blight, sheath blight, and rice blast, and fine-tuning a second MobileNetV2 [2] or EfficientNet-B0 on this paddy-enriched set.

R4 — PuLP MIP Activation for F4. Activating the already-coded PuLP mixed-integer programming formulation as the primary area-allocation solver, enabling inter-crop rotation constraints and multi-season planning.

R5 — Paddy Integration Across F2 and F4. Adding paddy as a sixth class to both the crop-recommendation stack and the price-prediction model, using the LightGBM production predictor [7] with 24 features.

R6 — End-to-End Deployment Study. Deploying ASICOP across instrumented farmer fields for one complete Maha or Yala cropping season to validate the integration contribution empirically and generate ground-truth telemetry for R2 and R3.

XIII. REFERENCES

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